MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION

MGEDSCCHE200-INORGANIC CHEMISTRY-1
MODEL QUESTION PAPER

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer All questions

Each question carries 1 marks

1. The Born-Haber cycle is used to calculate
   a) Ionisation energy
   b) Lattice energy
   c) Electron affinity
   d) Electron density (CO-1 , U)

2. Fajan's rule helps to predict
   a) Molecular Structure
   b) Hybridization
   c) Dipole moment
   d) Covalent character in ionic compounds (CO-1 , U)

3 According to VSEPR theory what is the structure of SF6

   a) Linear
   b) Octahedral
   c) Tetrahedral
   d) Square planar (CO-1, A)

4 Bond order of N_2 is

   a) 1
   b) 2
   c) 3
   d) 4 (CO-1, A)

5 The common oxidation state of lanthanides is

   a) +2
   b) +3
   c) +4
   d) +1 (CO-2,K)

6 The atom bomb is based on which nuclear process

a) Nuclear fusion
b) Nuclear Fission
c) Electron capture
d) Alpha decay (CO-3, U)

7 The half life of a radioactive isotope is

a) The time required for half of the radioactive nuclei to disintegrate
b) Time taken for the nucleus to emit all radiation
c) Time taken for nucleus to disintegrate completely
d) Time taken for energy to reduce to half (CO-3, U)

8 Give an example for ambidentate ligand

a) EDTA
b) SCN^-
c) NH_3
d) H_2O (CO-4, U)

9 Which is an example for weak field complex

a) CN^-
b) CO
c) NO_2^-
d) I^- (CO-4, U)

10 The color of coordination compounds is mainly due to

a) Charge transfer
b) Proton transfer
c) d-d transition
d) MLCT (CO-4, U)

[1x10 = 10]

Part B

Short Answer Type questions

Answer any 4 question

Each question carries 3 marks

11. State Fajan's Rule and explain its significance in determining the nature of bonding. (CO-1, A)

12. Predict the geometry and hybridization of XeF_2 using VSEPR theory. (CO-1, U)

13. What is lanthanide contraction? What are its consequences? (CO-2, U)

14. Differentiate between natural and artificial radioactivity . Give examples of each. (CO-3, U)

15, What are chelates ? Give examples. (CO-4, K)

16. Discuss the geometrical isomerism in Ma_3b_3 type complexes with examples. (CO-4, U)

[4x3 = 12]

Part C

Short Essay Type Questions

Answer any 2 question

Each question carries **7** marks

17. Describe how hybridization helps in explaining the geometry of XeF_4 and SF_6. (CO-1, U)

18 Discuss on the separation of lanthanides by Ion exchange method. (CO-3, U)

19 Explain how d orbital splitting in tetrahedral complexes differs from octahedral complexes. Give the splitting in both complexes. (CO-4, A)

[2x7 = 14]

Part D

Essay Type Questions

Answer any 1 question

Each question carries **14** marks

20 Give the difference between nuclear fission and nuclear fusion. Discuss on their advantages and dis advantages. (CO-3, U)

21 Discuss the characteristics of transition metal. Comment on how it differs from other elements in the periodic table. (CO-4, An)

[14x1 = 14]